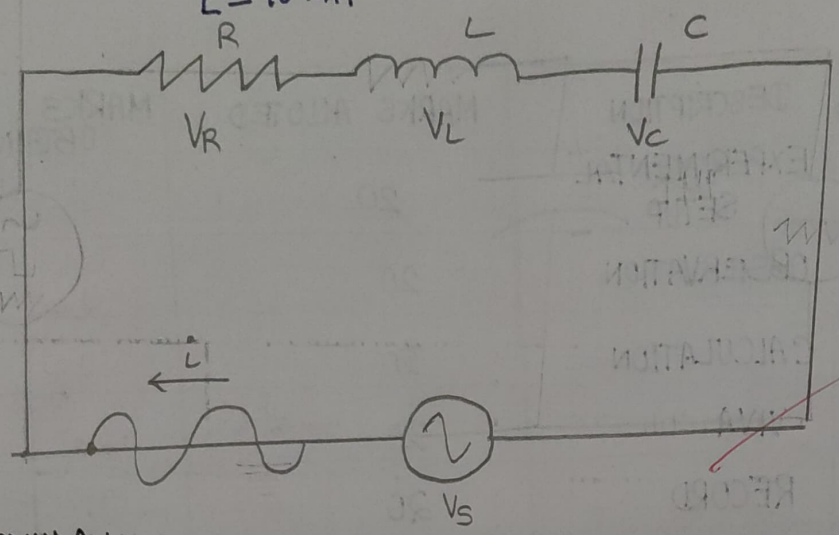
CIRCUIT DIAGRAM:-

choose $R = 10k\Omega$ and $C = 0.1\ \mu F$ (values of R and C can be changed).

$L = 10\ mH$



FORMULA :

During charging, Voltage across capacitor is given by.

EXP: 8
DATE: 17/03/22

TRANSIENT ANALYSIS OF SERIES ~~RC~~ RLC CIRCUIT

AIM:-

To determine transient response of a series ~~RC~~ RLC circuit using Multisim Simulator.

APPARATUS REQUIRED:-

S.No	Components	Specification	Quantity
1.	CRO	10 MHz	1
2.	DCB	–	1
3.	Resistors	10 kΩ	–
4.	Bread Board	–	1
5.	Connecting wires	–	–
6.	PC with Multisim Simulator	–	–

PROCEDURE:-

1. Make connection as per the circuit diagram.
2. Calculate the values of $V_c(t)$ and write in table. Compare the calculated values and measured values of voltage in both the cases.

TABULATION:-

(i) Voltage across Capacitor:-

V_{in} Amplitude:

V_{in} Frequency:

Time constant $\tau = RC =$

Time (t)	$V_c(t)$ - Charging Cycle (volts)		$V_c(t)$ - Discharging Cycle (volts)	
	Observed value	Calculated value	Observed value	Calculated value
$t=0$	0	0	V	V
$t=\tau$	0.63	0.7	0.368V	0.4
$t=2\tau$	0.865	0.95	0.135V	0.25
$t=3\tau$	0.950	1.0	0.05V	0.09
$t=4\tau$	0.982	1.1	0.01	0.02
$t=5\tau$	0.993V	1.0	0.007V	0.08

(ii) Voltage across Inductor:-

V_{in} Amplitude: 10V

V_{in} Frequency: 1kHz

Time Constant $\tau = L/R =$

O/P verified

Time (t)	$V_L(t)$ - Voltage across inductor
	Observed value
$t=0$	10 V
$t=\tau$	3.68V
$t=2\tau$	1.35V
$t=3\tau$	0.50V
$t=4\tau$	0.18V
$t=5\tau$	0.007V

(iii) Voltage across Resistor:-

Time(t)	$V_R(t)$ - Voltage across resistor
	Observed value
$t=0$	0
$t=\tau$	6.32
$t=2\tau$	8.65
$t=3\tau$	9.50
$t=4\tau$	9.82
$t=5\tau$	9.93

S.No	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	EXPERIMENTAL SETUP/ CONNECTION	20	20
2.	OBSERVATION / READINGS	20	20
3.	CALCULATIONS / RESULTS	20	20
4.	VIVA	20	20
5.	RECORD	20	20
		TOTAL	100

100

RESULT:-

Thus, the transient analysis of series RC circuit is done practically and verified with theoretical values using Multisim Simulator.